

Public Funds Treasury Dashboard Project Charter

Agile Software Engineering Capstone

This draft proposes a local treasury dashboard for a hypothetical public-sector Commerce Bank client. It sets a feasible semester boundary and identifies the decisions the team and professor need to confirm before approval.

Draft for team and instructor review. Confirm the marked names, dates, and team agreements before submission. Detailed requirements and sprint scheduling belong in separate documents.

Document field	Team entry
Project title	CivicFunds: Public Treasury Management and Cash-Flow Forecasting System
Team name	Merge Conflict
Team members	Efren Torres, Elias Damian, Kobe Lewis, Deep Toor
Project manager or team lead	Elias Damian
Project sponsor	Commerce Bank capstone mentors; specific approving mentor to confirm
Customer or client	Commerce Bank capstone (hypothetical public-sector client use case)
Instructor	Shah, Syed Jawad Hussain
Start and target end dates	September 28, 2026 to December 16, 2026 (final presentation)
Version status and date	0.1 Draft, September 25, 2026

Revision history

Version	Date	Author	Change and approval
0.1	September 25, 2026	Elias, Merge Conflict	Draft assembled from scope document and Commerce brief; awaiting team review
0.2	September 25, 2026	Elias, Merge Conflict	Added team names, project dates, and selected stretch goals; frontend recommendation pending team decision

1 Project Purpose and Authorization

This section explains why the project should exist and gives the team authority to proceed. Describe the problem before proposing detailed features.

1.1 Problem or opportunity

Public-sector finance directors must track cash across funds with different restrictions, scheduled obligations, and expected revenue. In a realistic treasury workflow, this information can be dispersed across reports and planning records, making the amount available for use difficult to assess quickly. A current cash total alone does not show what is restricted or already committed. That uncertainty can make near-term planning and early identification of low balances more difficult. This is a proposed use case, not a verified problem reported by Commerce Bank.

1.2 Project justification

Decision test	Team explanation	Evidence or source
Needed	A public-sector treasurer needs to distinguish cash on hand from restricted and committed money and anticipate near-term pressure. This is a plausible use case for the capstone.	Team Phase 1 scope; Commerce capstone brief requires a financial application
Feasible	Four developers can demonstrate one organization, four fund types, local mock data, and one organization-wide 30-day ML estimate between September 28 and December 16, 2026.	Phase 1 scope and team schedule assumption
Optimum	A focused prototype exercises the required full stack, finance workflow, authentication, database, frontend frameworks, and testing while avoiding live integrations.	Commerce requirements and Phase 1 exclusions

1.3 Product vision

For public-sector finance directors who need a clear view of available cash and upcoming obligations, the product is a treasury web application that combines fund visibility and a 30-day planning forecast.

1.4 Authority to proceed

Authorization item	Entry
Decision	Proposed; instructor and sponsor review pending
Conditions	Confirm frontend library choice, team roles, course checkpoints, and named reviewers. Selected stretch goals are self-hosted authentication with password reset and persistent sign-in across browser restarts.
Authority granted	Upon approval: finalize requirements and architecture, build mock data, and plan the first sprint.

2 Stakeholders and Team Authority

Identify everyone who can affect the project or is affected by its outcome. Record their needs and decision authority so the team knows whom to consult.

2.1 Stakeholder register

Stakeholder or group	Interest needs or expectations	Decision or contribution	Contact or review cadence
Commerce Bank capstone mentors	Financially relevant prototype and clear demonstration; no claim of production adoption	Give feedback on concept and major scope changes; approval authority to confirm	Mentor contact provided in brief; check-in and final presentation
Commerce Bank (academic customer)	Working full-stack finance application consistent with its brief	Provide requirements and feedback through mentors	Midsemester check-in; final presentation
Hypothetical finance director or treasurer	See usable cash, obligations, fund details, and forecast assumptions	Validate workflow in team-led scenario; no real user access assumed	Review with mentor/instructor at check-in
Professor Shah, Syed Jawad Hussain	Feasible charter, course compliance, tested deliverable	Clarify grading and approve course scope	Course channel and milestone reviews; confirm schedule
Four-person student team	Clear assignments, achievable scope, shared testing	Implement, document, test, present	Weekly planning and frequent check-ins; confirm
Project coordinator Becca Fantasma	Coordinate capstone communication and presentations	Logistical guidance; approval role to confirm	As needed through course contact

2.2 Team roles and responsibilities

Team member	Primary role	Charter responsibility	Decision authority or backup
Efren Torres	Database/Backend/ML	SQLite schema, Python ML prototype	Backup and decision area to confirm
Kobe Lewis	Frontend owner/ML	Dashboard, responsive UI, Bootstrap and chosen frontend library, Python ML prototype	Backup to confirm
Deep Toor	Backend and database owner/ML	PHP, registration and authentication, Python ML prototype	Backup to confirm

Team member	Primary role	Charter responsibility	Decision authority or backup
Deep	Project Manager/Frontend /ML	Coordinate scope wording and requirements draft, Python ML prototype	Backup to confirm

2.3 Decision and escalation rules

The four developers will make routine design decisions together and record them in the backlog. The project lead will settle schedule and backlog conflicts after hearing the team. Changes to the agreed scope, course requirement interpretations, or sponsor-facing commitments will be raised with the instructor and Commerce mentors before adoption. Confirm this decision rule with the team.

3 Goals Objectives and High Level Features

Goals describe the desired direction. Objectives make the expected outcomes specific and measurable. Features identify the main functions that deliver value to the client or users.

3.1 Project goals

Goal ID	Goal statement	Why it matters
G-01	Make a hypothetical public-sector organization's cash position understandable in one place	Supports planning across restricted cash, funds, and commitments
G-02	Demonstrate a reliable, secure, financially relevant full-stack application	Meets Commerce capstone learning and demonstration goals
G-03	Make a bounded forecast useful for early review of cash pressure	Shows expected activity and model estimates distinctly

3.2 Measurable objectives

Objective ID	Specific measurable objective	Target date	How the team will verify it
OBJ-01	Demonstrate registration, login, logout, profile/settings, dashboard and at least two additional working pages	December 16, 2026	End-to-end walkthrough and automated tests
OBJ-02	Produce a 30-day organization-wide forecast in weekly groupings with known amounts labeled separately from ML estimates	December 16, 2026	Forecast demo and evaluation on held-out mock historical data
OBJ-03	Pass tests for authentication, calculations, alerts, transaction filters and CSV export; complete two selected stretch goals	December 16, 2026	Automated test results and demonstrated stretch-goal evidence

3.3 High level features

Feature ID	Client valued feature	Primary user or stakeholder	Priority
F-01	Accounts, login and profile/settings	Finance director and Commerce reviewers	Must
F-02	Dashboard and four-fund drill-down with cash and transaction details	Finance director	Must
F-03	Planning records, alerts, CSV export and 30-day ML-assisted forecast	Finance director	Must
F-04	Self-hosted authentication with password reset; persistent sign-in across browser restarts	Commerce reviewers	Must

4 Scope Priorities and Deliverables

4.1 Scope boundaries

In scope	Out of scope
Local app with mock data; register/login, profile/settings, dashboard, fund detail, transactions, inflows/commitments, CSV, alerts and organization-wide 30-day forecast; required testing and two stretch goals.	Real bank integrations, actual customer records, payment initiation, production deployment, municipal ERP, automated financial decisions, per-fund ML and 90/120-day ML.
Proposed stack: HTML/CSS/JavaScript, Bootstrap, Vue (recommended; team decision pending), PHP, SQLite, Chart.js and Python/scikit-learn. Extended 90/120-day known-only views remain in Phase 1 scope.	No real money movement or card data; no external APIs without professor approval.

4.2 Project priorities

Rank the priorities from 1 to 3. Use each rank once, then explain the tradeoff rule.

Priority area	Rank	Team agreement
Scope and product value	3	Protect core user workflow; defer optional enhancements when capacity is tight
Schedule and course deadlines	1	Meet fixed course checkpoints; reduce nonrequired scope early
Quality and technical standards	2	Protect correct calculations, hashed passwords and critical tests before polish

4.3 Major deliverables and milestones

ID	Milestone or deliverable	Tangible evidence	Target date or sprint	Owner	Reviewer
M-01	Charter approved	Reviewer decision and approved charter	Sprint 0; starting September 28	Team lead to confirm	Instructor and mentor
M-02	Requirements baseline	Use cases, functional and quality requirements	Sprint 1; date to confirm	Team to assign	Instructor
M-03	Architecture reviewed	Architecture choice and integration proof	Sprint 2; date to confirm	Team to assign	Team and mentor
M-04	Midsemester increment	Login, sample dashboard, data and feedback notes	Course check-in date to confirm	Whole team	Commerce mentors

ID	Milestone or deliverable	Tangible evidence	Target date or sprint	Owner	Reviewer
M-05	Feature complete	Testable integrated release and evaluation	Penultimate sprint; date to confirm	Whole team	Team lead
M-06	Final delivery	App, tests, documents, presentation	December 16, 2026	Whole team	Instructor and Commerce mentors

5 Resources Constraints and Assumptions

5.1 Preliminary resource and budget estimate

Estimate resources at a level appropriate for initiation. Student labor can be recorded in person-hours even when no money changes hands.

Resource	Estimated quantity or effort	Estimated cost	Source or availability	Owner
Student effort	4 people × approximately 11 weeks × 6–8 hours/week = 264–352 person-hours (planning estimate)	\$0 course labor	Student availability to confirm	Team lead to confirm
Hosting or cloud services	Local demonstration only	\$0 anticipated	Team laptop / local runtime	Backend owner to confirm
Development tools	Editor, version control, PHP, SQLite, Python and selected libraries	\$0 anticipated	Free tools; versions to confirm	Whole team
Equipment or test devices	Four development laptops; browser testing	\$0 if existing	Student-owned equipment	Whole team
Mock data	Seeded historical transactions and planning records	\$0	Generated by team	Data owner to confirm

5.2 Constraints

Constraint ID	Limit placed on the product or project	Source	Effect on the team
C-01	Finish within course schedule	Course calendar	Defer optional features before required checkpoints
C-02	Use only mock financial and card data	Commerce brief and Phase 1 scope	No live bank connectivity or real money processing
C-03	Use CSS framework, additional frontend framework/library, password hashing, registration, testing and two stretch goals	Commerce capstone brief	Architecture and backlog must include each item
C-04	Run local Phase 1 without an external API	Team scope	Use local services and local ML inference

5.3 Assumptions and dependencies

ID	Type	Statement	How the team will validate or respond	Owner	Review date
A-01	Assumption	Four developers are available from September 28 to December 16, 2026, with 6–8 hours each weekly (estimate)	Confirm availability and deadline with team/instructor	Team lead to confirm	Sprint 0
D-01	Dependency	Consistent mock data for forecast and all financial views	Create seeded dataset; simplify model if history is insufficient	Data owner to confirm	Sprint 1
D-02	Dependency	Selected self-hosted password reset and persistent sign-in depend on safe account recovery and revocable, expiring remember-me tokens.	Design and test both workflows before feature completion; document security choices.	Team lead to confirm	Sprint 2

6 Success Criteria Risks and Agile Working Agreement

6.1 Project success criteria

Criterion ID	What success means	Measure or threshold	Evidence	Reviewer
SC-01	Primary user can complete core financial workflow	Register/login, view four funds, edit planning records, inspect forecast and alert, filter/export, logout	End-to-end demonstration	Instructor / mentors
SC-02	App meets core technical requirements	Hashed passwords, protected pages, selected frontend library and CSS framework; critical tests pass	Code and test evidence	Instructor
SC-03	Forecast and delivery are explainable	30-day weekly forecast labels known vs estimated values; ML measured on held-out mock data; the two selected authentication stretch goals shown	Demo, evaluation and presentation	Commerce mentors / instructor

6.2 Initial risks and obstacles

Risk ID	Potential problem	Likelihood	Impact	First response	Owner
R-01	ML data too sparse for useful forecast	High	Medium	Generate realistic history early, compare simple baseline, report limitations	ML owner to confirm
R-02	PHP and Python integration consumes time	Medium	High	Build a minimal end-to-end integration in Sprint 2	Backend/ML owners
R-03	Required pages or selected authentication stretch goals missed	Medium	High	Trace brief into backlog and check every sprint review	Team lead
R-04	Inconsistent balances across funds and forecasts	Medium	High	Centralize definitions and test calculations against known fixtures	Backend owner

6.3 Agile working agreement

Working practice	Team agreement
Sprint length and ceremonies	Proposed two-week sprints; planning at start, brief twice-weekly check-ins, demo and retrospective at end; confirm with team
Backlog ownership	Team lead keeps prioritized backlog; mentor and instructor feedback becomes reviewed backlog items
Definition of done	Peer review, working integration, relevant automated tests, updated notes and demonstrated acceptance criteria
Communication and response time	Team channel and attendance policy to confirm; identify blockers at next check-in
Scope change	Team estimates change; instructor and mentors review major boundary changes; record approval and update charter/backlog
Conflict or missed commitment	Discuss with owners promptly; reassign or reduce optional scope; escalate course-impacting issues to instructor

7 Approval and Submission Check

Approval confirms that the reviewers understand the project purpose, boundaries, expected outcomes, and initial authority. It does not freeze the entire backlog. Record approved charter changes in the revision history.

7.1 Review decisions

Reviewer	Role	Decision	Conditions or comments	Date
Commerce Bank mentor — name to confirm	Sponsor reviewer	Pending review	Confirm mentor authority and any conditions	To confirm
Shah, Syed Jawad Hussain	Instructor	Pending review	Confirm course dates and requirements	To confirm
Student project lead — name to confirm	Team lead	Pending team acceptance	Confirm roles, priorities and work agreement	To confirm

7.2 Signatures

Name and role	Signature	Date
Commerce mentor — name to confirm		
Professor Shah, Syed Jawad Hussain		
Student project lead — name to confirm		
Additional approver if required		

7.3 Final submission checklist

- ☐ The problem or opportunity is clear and does not begin with a predetermined solution.
- ☐ The charter explains why the project is needed, feasible, and a responsible use of resources.
- ☐ Stakeholders, team responsibilities, and decision authority are clear.
- ☐ Goals, measurable objectives, and high level features agree with one another.
- ☐ In scope and out of scope boundaries can guide backlog decisions.
- ☐ Milestones name tangible deliverables, target dates, owners, and reviewers.
- ☐ Constraints, assumptions, dependencies, success criteria, and initial risks are recorded.
- ☐ The agile working agreement explains how the team will plan, review, test, communicate, and handle changes.
- ☐ Every bracketed prompt has been replaced or removed before submission.
- ☐ The version, date, author list, and approval decisions are current.

Draft filename: Commerce_Treasury_Project_Charter_Draft.docx